The Technical Challenges of the SHIELD Act on Small Businesses

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Abstract

Small businesses that are handing New York State resident data are expected to comply with the Stop Hacks and Improve Electronic Data Security Act or the SHIELD Act. Although they are expected to comply with this act there is limited research that is available to show how they are achieving these requirements. The purpose of this proposed study is to determine how small businesses approach compliance with the SHILED Act. This study focuses on awareness, implementation methods, resource constraints, and internal or external IT support. These findings can be achieve though the quantitative survey that is proposed to collect this information to determine how small businesses are complying with the SHIELD Act. Since this study is conceptual, expected results are calculated by existing literature and industry reports. These are used to determine the expected results of small businesses filling out the survey. While offering insight to inform both small businesses and policy makers and future research based on this area.

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Introduction

On July 25, 2019, The state of New York signed into law the Stop Hacks and Improve Electronic Data Security Act or the SHIELD Act (State, 2019). This act aims to strengthen the protection of New York State residents' data-security within businesses. This Act redefined and expanded what is private information should be kept secure. This Act expanded private information to include social security numbers, driver's licenses numbers, account numbers, biometric information, username, email addresses, and password credentials (State, 2019). With this expanded list of protected private information businesses were expected to include the necessary safeguards to protect this information.

In this topic there is little to no information in regard to the technical, financial, and operational challenges that small businesses face when complying with the SHIELD Act. Small businesses may not have the resources or are struggling to comply and adhere to this new form of regulation. These small businesses may be able to benefit from knowing how other small businesses are achieving these requirements. This can be valuable information to determine how on track small businesses are complying with this law.

Some questions that can be answered are:

Are small businesses outsourcing their technical requirements? Are security controls difficult for them to implement? Do they have the budget for these changes? Are there areas that they need to look at that they didn't have to before? Were they already protecting this information? Is this information data they were already collecting?

Goals:

- **Identify Barriers affecting compliance**
- **Determine different compliance approaches**
- **Compare these different compliance approaches**
- **Determine the main way small businesses are complying**
- **Allow small businesses to see how they are comparing among their peers**
- **How small businesses are allocating resources to cybersecurity**
- **Any unknown difficulties that are arising with complying with this law**

Objectives:

- **Document security controls that are being used**
- **How confident are small businesses in complying with this law**
- **Comparison of small businesses**
- **External support**
- **Internal support**
- **Resource constraints**

Literature Review

Existing literature on the New York State SHIELD Act primarily examines regulation requirements and impact on organizations that do not adhere to compliance requirements. Many of the articles that are out there focus on the impact of this act on business in New York state and nationwide. This is similar to other state laws that cover private data protection for residents. This act looks to define protected data and what business needs to do to enforce this law. Current literature provides clear indications of what is protected and unprotected under this act, although there is limited information on what counts as necessary safeguards to implement. .

Research shows that the SHIELD Act, "… establishes new minimum security requirements for all persons and entities that possess computerized private information about New York residents," including "technical, physical and administrative safeguards" (Journal, State's SHIELD Act in effect to regulate cybersecurity protections, 2019). The SHIELD Act sets up the minimum-security requirements that are needed to protect private information. With these new security requirements businesses are being forced to make changes and re-evaluate different areas of protected information. Data that is being kept in business now needs to follow the required minimum-security requirements set by the SHIELD Act. This law is also starting to affect new technological advancements.

One particular area that is discussed is new AI driven technology. It is not just the SHIELD Act that has affected this new technology. According to one of the articles, other acts such as the Health Insurance Portability and Accountability Act, the Fair Credit Reporting Act, and the Uniform Trade Secrets Act and the Defend Trade Secrets Act are also affecting new AI models (Journal, Evolving data security and privacy considerations for businesses, 2023). These data protection acts are deciding how information can be stored and processed through these AI models. If the correct security requirements are not met when implementing AI models this can lead to violating any of these mentioned acts. To ensure these acts are being followed all articles discuss how internal audits should be conducted to make sure all aspects of the business are following the security requirements of each act. Along with following the act in new technology, companies need to make sure all New York state data is being protected.

This act not only protects data that is being stored by New York State companies but any data that is being stored about any resident of the state. Regardless of the location of the business organizations need to follow this Act if they are dealing with any New York State information that is protected (Mitchell, 2019). The minimum-security requirements are needed to be met when it comes to business dealing with this protected information. Reasonable security controls must be implemented by the business no matter its location (Payne, 2020). This act forces business nationwide to follow these requirements to protect information about New York State residents. These articles cover a lot of how the Shield Act requires businesses to implement security measures to follow. But there are some areas that can have further research be done.

Companies that do not adhere to following this act can face legal repercussions from the Attorney General of New York state. In my research I have found multiple cases of larger businesses being prosecuted by the state of New York. This can become a major concern for small businesses that are also expected to follow the SHIELD act. These cases were found on the Attorney general of New Yorks website.

There have been multiple cases that have been brought to the supreme court of the state of New York that have involved businesses not following the guidelines in the SHIELD Act. Companies that the State of New York has filed against are National General Holding Corp., Dunkin Donuts, National Amusement Assurance of Discontinuance, Capital One and CITI Bank. These companies have violated this act by not implementing enough safeguards to protect private information of New York residents. Each of these businesses has had to face legal action because of their failure to comply with this Act.

Multiple settlements and fines have been paid out by businesses because of complaints that have been filed against them. In the case of State of New York v. Dunkin, Dunkin was forced to settle on $650,000 in penalties (The People of the State of New York, by Letitia James, Attorney General of the State of New York, v. Dunkin Donuts, 2019). This is not the only time that a business was forced to pay fees for violations. In State of New York v. National Amusements, they were charged a penalty of $250,000 to the State of New York (York, 2024). Although National Amusement did not admit or deny any wrongdoing they were required to pay this

penalty. This can be easy for a larger business to pay for but can be costly for a small business to pay.

Smaller businesses must be more careful to adhere to the safeguards required by the SHIELD Act. Since if they were to face one of these penalties, they may face financial hardship in paying. Larger businesses are more able to pay their penalties in the event a breach happens because they have more financial liquidity. This is not to say that they will face paying the fines instead of following the SHIELD Act. This is more so because they can financially recover from this worst-case scenario. Financial penalties are not the only thing that businesses that do not comply can be faced with. They can also face Mandatory reforms of their security safeguards.

Mandatory reforms of a business can be difficult for small businesses to achieve through their limited resources. So, it is important for them to seek external and internal resources to comply with the necessary safeguards for this act. This continues my question of how small businesses are complying with the SHIELD Act. What necessary safeguards are they implementing to avoid violating this act? As well as are there any financial constraints that they may be facing?

Businesses that have been filed against have also found themselves unprepared for any security event. Such as in the case of State of New York v. National General, although this case has yet to be ruled on, they are facing action against not having a proper security plan. In this case National General failed to notify customers and regulators of the breach that happened in 2020 and 2021which affected 155,000 New Yorkers (State of New York v. National General Holdings Corp. et al., 2025). One of the necessary thing's businesses must do under this act is notify affected parties of their data being stolen. This case can indicate that some businesses are only complying once they are being forced to comply.

Facing penalties and reforms can help to persuade other businesses to implement the necessary safeguards before facing the same legal actions. With multiple cases being brought to the supreme court of New York State will show that The SHIELD act is being enforced. Forcing businesses that may not have thought about the SHIELD act to start thinking about it. Small businesses are hopefully already seen the repercussions that larger businesses are facing and implementing any safeguards that they may be missing. This is important not just for small businesses to consider but all businesses that handle New York state resident data.

Businesses have had clear and consistent repercussions for violating the terms of this act. Whether that would be financial penalties or force compliance reform. Small businesses should be making sure they are taking the necessary steps to implement the correct safeguards to protect private information of New York State Residents. This can be either by external resources or allocating internal resources to comply with protecting this information.

With all this information out there, that is discussed about the SHIELD Act, there are multiple areas that are not being discussed. One area such as the effect this Act has on small business. There is no information that I was able to find in these articles that explores the financial, technical or operational challenges in small businesses. Areas such as resource limitation, third-party management, and government support are not addressed within these articles. These areas can be discussed further in future studies to help determine the challenges that small businesses are facing when trying to comply with the SHIELD Act.

Conceptual Framework

For this research the proposed conceptual framework is documented in figure 1. The framework identifies each carriable for this study to look at. In this framework the level of compliance with the SHIELD Act is the dependent variable. This shows if small businesses are following the SHIELD Act's requirements. The independent variable of external and internal methods will change depending on how the small businesses are complying with the SHIELD Act. External pressures such as legal action can determine if small businesses are feeling pressured to follow this Act. Financial and technical resources can either limit or improve what external or internal methods are being used. While implemented safeguards are the mediating variable between compliance and methods that are used to achieve compliance. Finally, the control of this research is small businesses.

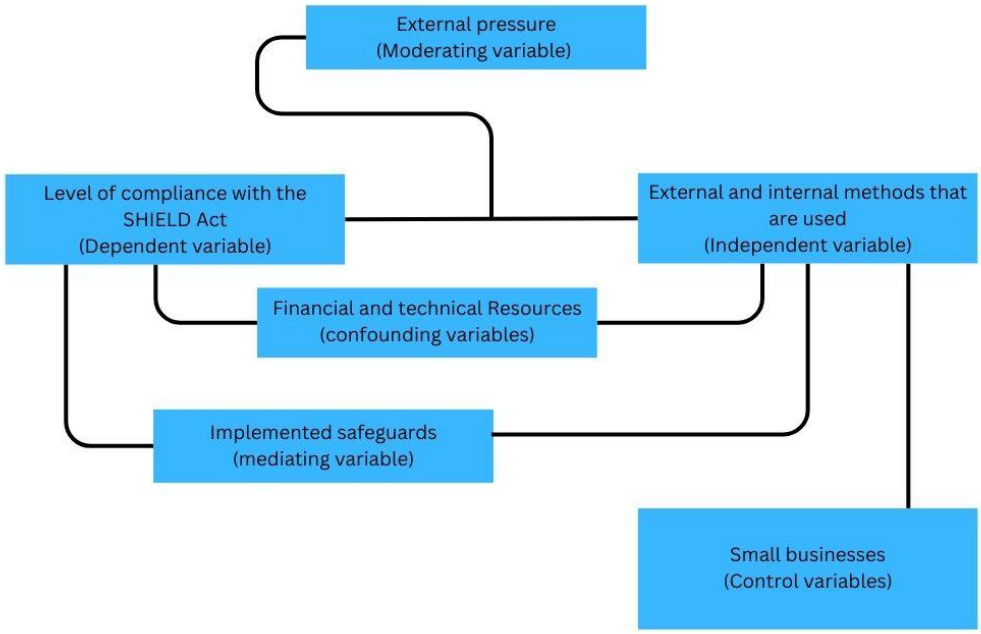

Figure 1. Conceptual Framework

Research Method

Participants

The Participants of this survey questionnaire will be small businesses that deal with New York State resident data that is protected under the NYS SHIELD Act. According to the PDF, "A population is a group of individuals that have the same characteristic(s)" (nova, 2025). So, for this project the population size will be small businesses that have less than 500 employees. This survey will ideally be filled out by the manager of data-security or the owner of the small business. Other information such as location, industry and years of operation will be collected in the sample group of this population.

Instruments

Data will be collected by a survey questionnaire to measure awareness, understanding and implementation methods. Small businesses will answer based on information pertaining to the SHIELD Act. This will include items such as risk assessments, employee training procedures, breach responses, and any compliance issues that may occur.

Procedures

To make sure this research can be replicated the following procedure will be taken. IRB approval will be acquired to ensure the questionnaire remains ethical in its findings. Then once this is completed the questionnaire will be distributed on small business networks online to obtain results. This survey will take 10-15 minutes for participants to fill out. This will be up for 6 weeks where "no intervention or treatment" will occur (nova, 2025).

Data Analysis

A quantitative research approach will be taken with this questionnaire. Data will be examined based on level of SHIELD Act understanding, safeguards implemented, and employee training. This data will be used to determine how small businesses are complying with the required minimum safeguards required for protecting New York State Residents data from breaches. Data may be also used to discover if there is a relation to business size and whether security measures are in-house or provided by a third party. An open-ended response will be displayed separately from data that is quantitative data if used.

Expected Results

Based on research from outside sources. I can predict the potential findings that may be seen when executing this survey. According to the 2022 (ISC)2 Cybersecurity Workforce study, "48% of organizations are choosing to outsource security services as their preferred approach to enable the maximum benefits…" (Larson, 2025). So, it is most likely that roughly 48% of small businesses are using third-party vendors to conduct safeguards and monitor their systems from

data breaches. According to other research done by (ISC)2 Cybersecurity Workforce Study conducted in 2024, businesses are also seeing pressure from the current reshaping of the cybersecurity landscape, such as the rise of AI and economic pressures. These factors can also lead to why they may have even higher records of small businesses choosing to use third-party vendors.

Small businesses on average are more likely to face data breaches and cyber attacks. According to the business impact report of 2023, 73% of small businesses have reported experiencing attacks within the previous year (Rinaldi, The Cost of Cybersecurity and How to Budget for It, 2026). While this is not the only reason this could show signs that either financial or lack of awareness can be present when conducting this survey. Small businesses could be seen spending low amounts of their budgets on cybersecurity. 59% of small businesses are seen to spend less than 10 hours on cyber security activities and 74% of them are seen to spend less than 10% of their budget on cybersecurity (Toomey, 2025). So, if small businesses are already seen spending less than 10% of their budget on Cybersecurity practices, this could indicate also the possibility of the lack of the necessary safeguards required by the SHIELD Act.

With lack of expenses and a high percentage of attacks on small businesses, it is likely to assume awareness is also lacking in small businesses. This can lead to the findings of the survey indicating that small businesses are less aware of the required necessary safeguards of the SHIELD Act. 64% Small businesses surveyed in 2025 by Coalition did not think that they were in active danger of threats or attacks even though this survey saw 79% of them have experienced one. (Toomey, 2025). Trends in other surveys would indicate that small businesses would still not have awareness of the SHIELD Act. This information would imply that small businesses could face legal action if any protected New York State Resident data is breached in one of these threats or attacks.

Discussion

With this information it is also safe to assume that little to no safeguards could be seen being impended. This could also indicate that the person taking the survey may not be aware of all the safeguards that are in place given the lack of awareness that it is being seen throughout other

surveys. It is also seen that many small businesses are using outside vendors for data storage. According to trends seen by SQ Magazine in 2026 60% of data is stored in the cloud unlike in 2015 in which only 15% of data was stored in the cloud (Ramirez, 2026). This could also increase the number of safeguards that could be seen from small businesses, which could show that small businesses are complying with the necessary safeguards of the SHIELD Act. Small businesses are seen to not have a vast knowledge of requirements that could be affecting them. Even with this information small businesses could be implementing some of the necessary safeguards of the SHIELD Act but are not meeting all requirements.

Conclusion

Overall, trends suggest that small businesses are facing a significant lack of cybersecurity safeguards and awareness that could affect their understanding of the SHIELD Act. High attacks, low budget and lack of awareness create the potential for small businesses to face potential legal action from the State of New York. Larger businesses have been seen to have legal actions taken against them. This has resulted in multiple settlements that businesses have had to pay out. Small businesses could face this identical scenario since they are seen having this lack of awareness and implemented safeguards required by the SHIELD Act. These findings also highlight the need and the importance of education and support for small businesses. This support and education can help prevent legal actions that could be caused by different Acts than just the SHIELD Act of New York State.

Limitation of Research

The primary limitation of this study is its reliance on secondary sources to conduct the anticipated results of the proposed survey located in Appendix A. In the absence of primary survey, data academic sources, industrial data, and cybersecurity best practice studies were used to gauge what the outcomes could be when conducting the actual survey. Due to the fact research in this area is limited, many of the sources are primarily related to cyber threats and attacks. Outside sources were not restricted to just sources involving the SHIELD Act.

Additionally, the SHIELD Act itself does not state the exact safeguards that New York State requires to be in place to protect New York State Resident Data. As a result, research questions and anticipated results were broadly related to safeguards for other cybersecurity regulatory actions. Although there can exist similarities between these and the SHIELD Act, differences can be present that could affect the accuracy of the inferred safeguards. Both in the survey that has been drafted and the information presented in the anticipated results.

Recommendations

Through the anticipated results of this study, recommendations can be made for small businesses, policy makers, and future research related to the New York State SHIELD Act.

Firstly, small businesses can become more aware of regulatory measures that may affect them. Common trends in research data show that small businesses are facing threats and attacks on a massive scale. Although many small businesses are seen to experience this, very few see themselves as a target for these to happen. Small businesses could benefit from continued education of recommended best practices and safeguards to protect their information.

As well as small businesses, policy makers can provide clear guidance and support for small businesses. The SHIELD Act requires the necessary safeguards to be implemented but does not specify what qualifies as the necessary safeguards that should be used. Providing detailed ways that Businesses are expected to protect New York State Resident data can help to educate small businesses on what they should be aware of.

Lastly, future research should focus on conducting empirical studies that gather the primary data from small businesses. It should include small businesses from a variety of industries. Survey based research such as the one made for this study should be used to collect this data, collecting this data can help further the understanding of how aware small businesses are of regulatory actions such as the SHIELD Act. While also discovering their budget and what types of safeguards they are using to secure protected data.

Appendix A

This appendix provides the written survey instrument that can be used to collect the quantitative research to examine small business awareness and compliance with New York State Stop Hacks and Improve Electronic Data Security (SHIELD) Act.

Survey questions

1. What is your business size?
    a) 1 – 49 employees
    b) 50 – 99 employees
    c) 100 – 149 employees
    d) 150 – 259 employees
    e) 250 – 500 employees
2. Which industry describes your business?
    a) Retail
    b) Insurance
    c) Medical
    d) Technology
    e) Non-profit
    f) Banking
    g) Other____
3. How long have you been operating as a business?
    a) Less than a year
    b) 1 – 5 years
    c) 6 – 10 years
    d) 11 – 15 years
    e) 15+ years
4. How would you rate your understanding of the data security requirements of the NYS SHIELD Act?
    a) No Understanding
    b) Limited Understanding
    c) Moderate Understanding
    d) Good Understanding
    e) Exceptional Understanding
5. Where did you first hear/learn of the SHIELD Act and its data protection requirements? (Select all that Apply.)
    - Government Website

- Legal Counsel
- IT department / Third party IT vender
- News or Media
- Professional Contact
- None Above

6. Who is the primary source for data security safeguards and implementing them in your organization?
    a) Internal IT Staff
    b) Third-party IT/Vendor
    c) Business owner/Manager
    d) Shared between internal staff and vendors
    e) Not assigned
7. Our security personnel, whether internal or external, actively manage our data security (e.g., updates, backups, security requirements)
    a) Strongly Disagree
    b) Disagree
    c) Neutral
    d) Agree
    e) Strongly Agree
8. Approximately How much money does your organization spend annually on IT or Cybersecurity?
    a) $0 - $500
    b) $501 - $2,000
    c) $2,000 - $5,000
    d) $5,000 - $10,000
    e) More than $10,000
    f) Not sure
9. (Physical Requirements) Devices that contain Private information protected under the SHIELD Act are secured when not in use.
    a) Strongly Disagree
    b) Disagree
    c) Neutral
    d) Agree
    e) Strongly Agree
10. (Technical Requirements) We limit private information store to what is necessary to complete required business operations
    a) Strongly Disagree
    b) Disagree
    c) Neutral
    d) Agree

e) Strongly Agree

11. (Technical Requirements) What tool(s) do your business use to prevent malicious emails (select all that apply)
    - Employee phishing training
    - Encrypted emails
    - Filters to detect and block potential malicious emails
    - Banner Warnings
    - AI-Detection methods
    - Other___
    - Unsure/not used
12. (Technical Requirements) Which of the following best describes how computers and devices are protected in your business?
    a) Antivirus/anti-malware software is used on all devices
    b) Some devices are protected by antivirus/anti-malware
    c) Built-in protectors are used on devices (e.g., Windows Defender, BitLocker, Factory Reset Protection, ETC.)
    d) Unsure
13. (Technical Requirements) When is sensitive data encrypted in your business? (Select all that apply)
    - When data is stored on computers or servers
    - When transmitted over the internet
    - When stored on mobile devices
    - We do not use encryption
    - Unsure
14. (Technical Requirements) Does your business monitor the system for suspicious or unauthorized activity?
    a) Yes, automated monitoring tolls are used
    b) Yes, reviewed by IT staff
    c) The third-party vendor monitors
    d) No Monitoring is being done
    e) Unsure
15. Where does your business store sensitive data? (select all that apply)
    - Local computer
    - Cloud Storage (e.g., OneDrive, Google Drive, etc.)
    - On-site Server
    - Third-party software platform
    - We do not collect sensitive data
    - Unsure
16. (Optional) Is there any information or other safeguards that you have implemented that were not covered?

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